# Abdulqader Alshihabi

**Software Engineer | Backend & AI Engineering**

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## SUMMARY

Computer Science student graduating in January 2027 (GPA: 4.62 / 5.00), with practical experience developing scalable backend systems, full-stack applications, and AI-powered computer vision and data processing solutions. Experienced in Python, PyTorch, Django, React, TypeScript, and PostgreSQL, with hands-on contributions across enterprise ERP workflows, deep learning vision models, semantic vector search, and institutional research platforms.

## TECHNICAL SKILLS

**Programming Languages:** Python, TypeScript, JavaScript, SQL, Java, Go

**Backend & Cloud:** Django, Django REST Framework, FastAPI, RESTful APIs, Role-Based Access Control (RBAC), Supabase, Docker, Azure

**Frontend Development:** React, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS, Canvas API

**Databases & Vector Data:** PostgreSQL, pgvector, PL/pgSQL, Relational Schema Design, Query Optimization, Vector Embeddings

**AI, Machine Learning & Vision:** PyTorch, YOLO11, Faster R-CNN, Computer Vision, ONNX / ONNX Runtime, Sentence-Transformers, Scikit-learn, LLMs (Ollama)

**Core Methodologies & Tools:** Git, GitHub Actions (CI/CD), Unit Testing, SDLC, Agile/Scrum, Linux, Postman, WebSockets

## WORK EXPERIENCE

**Research & Innovation Consultant (Part-time)** **Jul 2026 – Present**

*Institute of Research and Consulting Studies, Islamic University of Madinah | Madinah, Saudi Arabia*

- Engineered a locally hosted LLM extraction pipeline using Python, Ollama, and Qwen, processing 25,000+ unstructured academic research papers from the university repository, OpenAlex, and Scopus with zero cloud API costs.
- Accelerated comprehensive research data extraction from an estimated several months of manual review to just 10 days, implementing data validation, Arabic text normalization, and retry mechanisms to improve extraction reliability.
- Organized researcher profiles and domain expertise by generating dense vector representations with multilingual Sentence-Transformers and clustering techniques, designing optimized PostgreSQL schemas for relational and vector queries.
- Collaborated with the platform engineer to deliver research analytics for the commercialization engine and prepared technical progress reports and milestone presentations for university leadership.

**Software Engineer Intern** **Jun 2026 – Jul 2026**

*Asaqefah for Business Development | Saaed HR & Payroll ERP | Madinah, Saudi Arabia*

- Contributed to Saaed, an enterprise HR and Payroll ERP serving 30+ employees, developed to replace disconnected departmental spreadsheets with integrated transactional models.
- Developed core backend services with Django REST Framework and interactive frontend views with React and TypeScript on a PostgreSQL database.
- Accelerated payroll approval and finance reconciliation workflows by over 4x by automating the transfer of validated payroll calculations directly into Finance general ledgers.
- Implemented role-based access control (RBAC) and strict server-side validation rules to secure sensitive employee records and maintain transactional consistency across cross-departmental workflows.

## PROJECTS

**SiteSight – Edge AI Construction Safety & PPE Computer Vision** **May 2025**

github.com/Abdulqader-Alshihabi/SiteSight-Safety-Vision | *Technologies: YOLO11, PyTorch, ONNX Runtime Web, JavaScript, Canvas API, WebAssembly*

- Engineered an in-browser edge computer vision system running real-time YOLO11 inference (1x3x640x640 input tensor) to detect PPE safety compliance without server GPU infrastructure.
- Implemented client-side letterbox padding, multi-class confidence filtering, and non-maximum suppression (NMS) over 8,400 predictions to log hardhat and vest compliance directly on HTML5 Canvas.

**Faster R-CNN Deep Object Detection Pipeline** **Aug 2025**

github.com/Abdulqader-Alshihabi/Faster-RCNN-Object-Detection | *Technologies: PyTorch, ResNet-50 FPN, TorchVision, ONNX, FastAPI, PASCAL VOC, JavaScript*

- Built and trained a two-stage Faster R-CNN detector with a ResNet-50 Feature Pyramid Network (FPN) backbone across 20 object classes on 4,952 test images, achieving 24ms inference latency.
- Engineered multi-file ONNX weight graph chunking for client-side model execution and deployed a companion FastAPI prediction service with automated validation suites.

**Islamic University CRIS – Research Information System** **Oct 2025**

github.com/Abdulqader-Alshihabi/Islamic-University-CRIS | *Technologies: React, TypeScript, PostgreSQL, PL/pgSQL, Supabase, Scopus API, OpenAlex API*

- Centralized and reconciled 7,000+ academic publications and 70,000+ verified citations across Scopus and OpenAlex APIs, resolving fragmented institutional research records.
- Designed automated ETL pipelines in PostgreSQL and PL/pgSQL with multi-field deduplication algorithms, improving the speed and accuracy of faculty post-publication award evaluations.

**res-handler – AI Semantic Search & Research Matcher** **Sep 2025**

github.com/Abdulqader-Alshihabi/res-handler | *Technologies: Python, Go, PostgreSQL, pgvector, Azure OpenAI Embeddings, Docker*

- Engineered an AI semantic vector search service converting problem queries into 1,536-dimensional embeddings, indexing 5,000+ academic papers to match faculty expertise.
- Implemented cosine distance similarity lookup with pgvector over dense representations, achieving sub-200ms query retrieval latency across high-dimensional embeddings.

**Shakkel Learn – Corporate Learning Management System** **Nov 2025**

learn.shakkel.store | *Technologies: React, TypeScript, Supabase, Tailwind CSS, PostgreSQL RLS*

- Engineered a multi-tenant corporate learning platform supporting 4 distinct user roles with bilingual Arabic/English interfaces, automated assessments, and verified completion tracking.
- Configured 15+ PostgreSQL Row-Level Security (RLS) policies to enforce tenant data isolation and role-based access across corporate learning environments.

## EDUCATION

**Bachelor of Science in Computer Science** **Expected Graduation: January 2027 | GPA: 4.62 / 5.00**

*Islamic University of Madinah | Madinah, Saudi Arabia*

**Relevant Coursework:** Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Operating Systems, Computer Networks

## CERTIFICATIONS

- **Advanced Artificial Intelligence (Computer Vision) –** KAUST Academy & Universidad Politécnica de Madrid (2026)
- **Introduction to Artificial Intelligence (with Distinction) –** KAUST Academy & Universidad Politécnica de Madrid (2026)
- **KAUST Academy AI Specialization Selection –** KAUST Academy & Islamic University of Madinah (2026)
- **Asaqefah SandBox Innovation Program –** Asaqefah & Amazon Web Services (AWS) (2026)
- **Introductory Artificial Intelligence Program –** KAUST Academy & NTDP (2025)
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization –** DeepLearning.AI (2024)
- **Mathematics for Machine Learning: Calculus, Linear Algebra, Probability & Statistics –** DeepLearning.AI (2024)